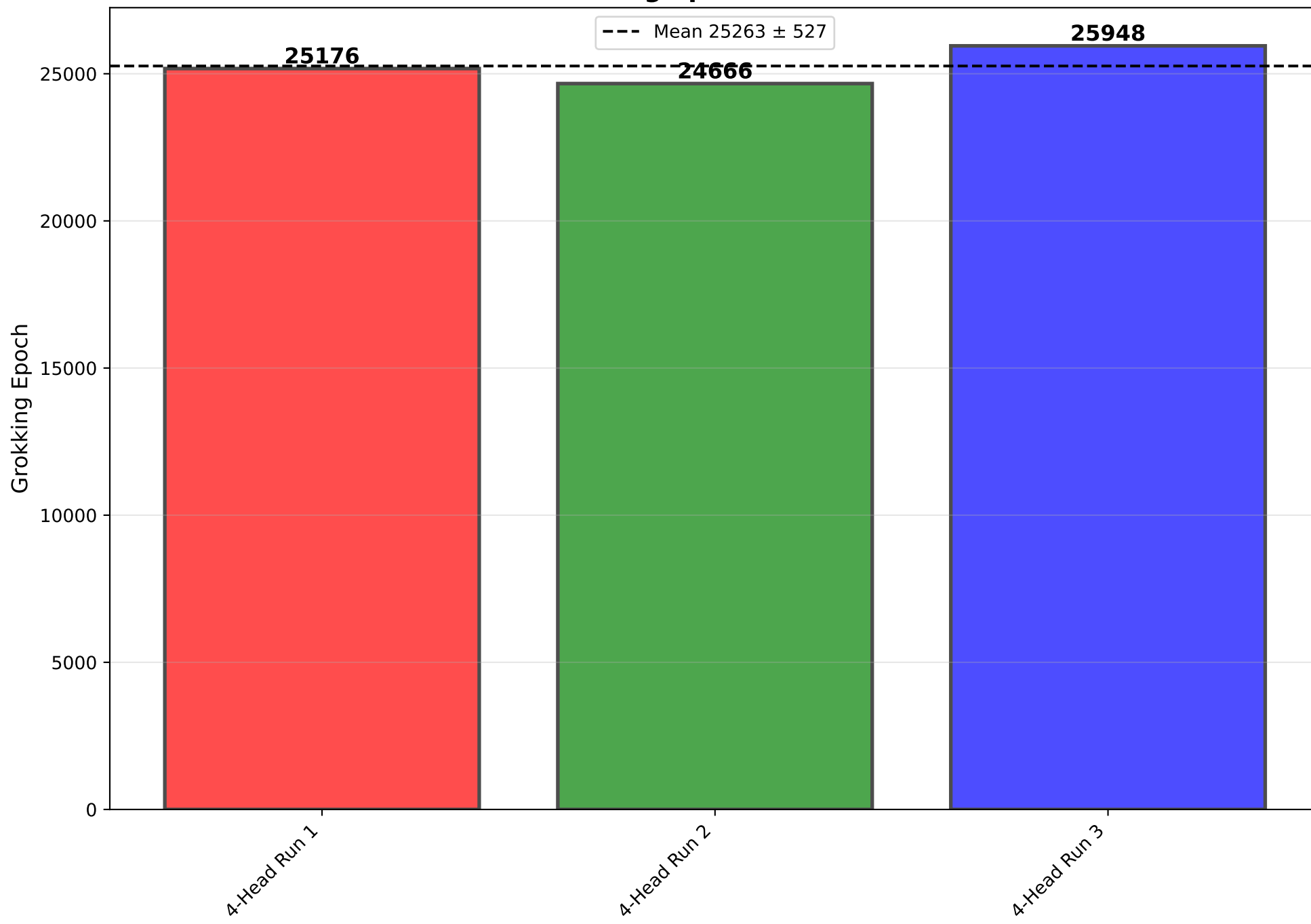
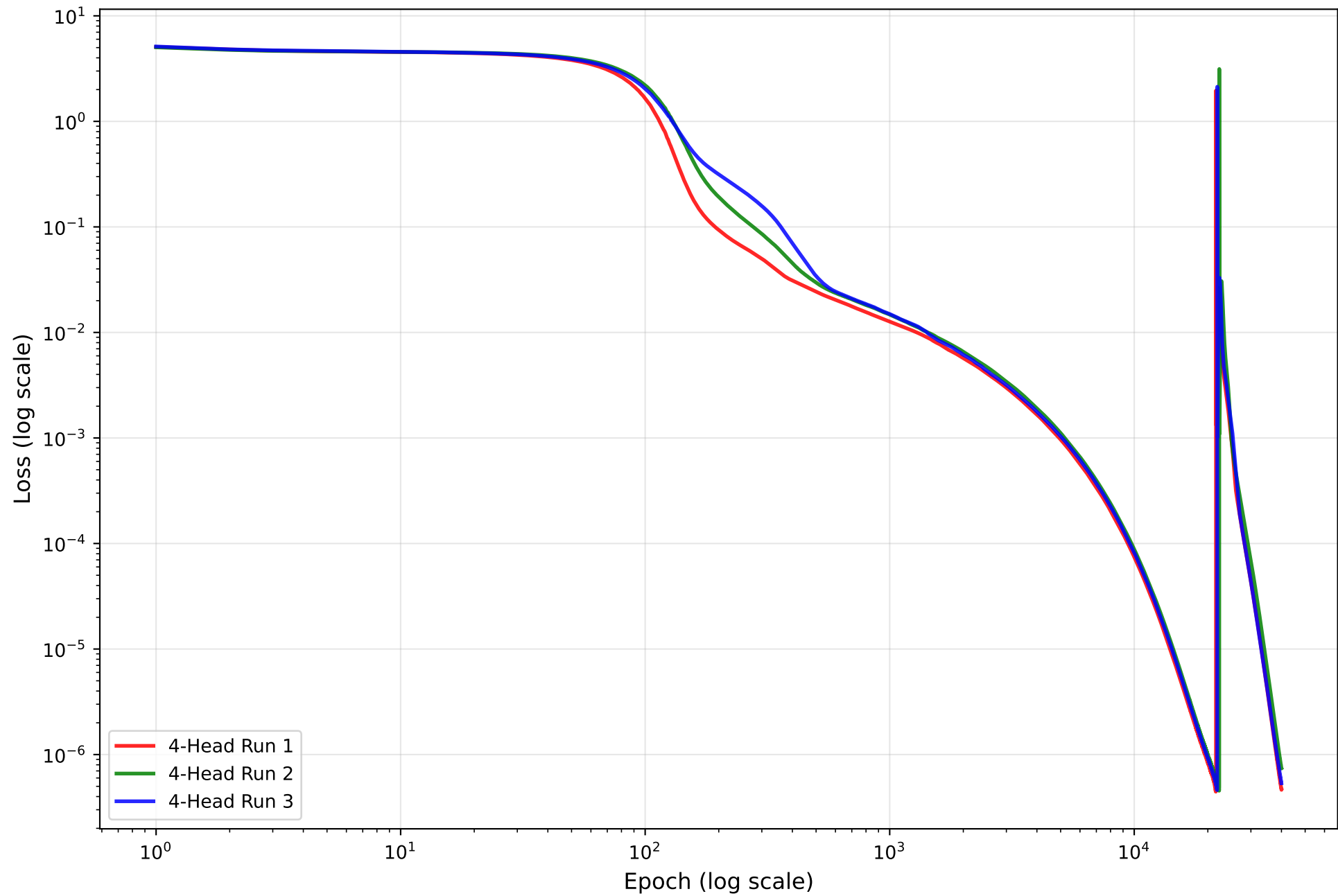


Four-Head Grokking Epoch Across Seeded Runs



Four-Head Loss Curves



FOUR-HEAD RUN CONSISTENCY ANALYSIS

Grokking Epochs (3 seeded runs):

Run 1: 25176 epochs

Run 2: 24666 epochs

Run 3: 25948 epochs

Mean: 25263.3 $\pm$ 527.0 epochs

Consistency: Moderate

Interpretation:

- Low std dev: predictor signal consistent across random seeds
- High std dev: predictor signal sensitive to initialization

Next Steps:

1. Validate L2 Norm & Dropout against the 3-criteria protocol
2. If both pass: move to the Spectral predictor
3. Build the baseline benchmark with the validated predictors